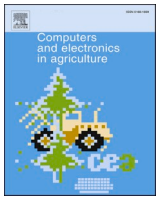
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Collaborative path planning and task allocation for multiple agricultural machines

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ABSTRACT

Path planning and task allocation are critical concerns in multi-machine collaborative operations for unmanned farms. Nevertheless, several problems remain in the operation of agricultural machinery, such as the slow path planning algorithm, the omission of the working area, and the unreasonable scheduling of machines, resulting in low efficiency and wasted resources. Collaborative and complete coverage path planning was achieved to solve the problems of slow path planning algorithms and the omission of working areas. The farm's electronic map was constructed using the topological map method. The improved Dijkstra algorithm based on priority queues was combined with three different complete coverage methods: the nested method, the reciprocating method, and the combination of nested and internal spiral path methods. The simulation results show that the improved Dijkstra method based on priority queues can effectively minimize the running time of the algorithm. The reciprocating method has a higher coverage index than the other two methods, with an average coverage rate of 94.73 %. To solve the problem of illogical scheduling of the same type of agricultural machines, an improved ant colony method was presented based on the whole working path to minimize the path cost. The simulation results show that the proposed method can allocate the task properly, and the path cost is reduced by 14 %–33 %. By combining the proposed path planning and task allocation methods, the whole-process path planning of a single agricultural machine and multiple agricultural machines of the same type was achieved, providing a technical solution for promoting the construction of unmanned farms.

1. Introduction

1.1. Unmanned farm

Smart agriculture is an advanced form of modern agriculture, and the unmanned farm is an important way to achieve intelligent agriculture. With the rise of a new generation of information technologies such as cloud computing, big data, the Internet of Things, artificial intelligence, and the deep integration of agricultural technology and information technology, the rapid development of digital agriculture and intelligent agriculture is being promoted (Sharma et al., 2022; Tao et al., 2021). An unmanned farm is an important approach to promoting the development of intelligent agriculture. It's through the remote control of farm facilities, equipment, machinery, etc., or the autonomous decision-making and tasks of intelligent equipment and robots. It's possible to realize uninterrupted production and operation of the unmanned farms in all-weather, whole-process, and whole-space (Lan et al., 2021; Li & Li, 2020). The collaborative operation of multiple agricultural machines of the same type (CO-MAMSY) in the field has become the focus of unmanned farm research (Wang, 2019; Li et al., 2019). The CO-MAMSY has the advantage of group collaboration and has great superiority in terms of task applicability, economy, and scalability, which is more conducive to the realization of autonomous operation in large-scale farmland. It mainly emphasizes the technology of multi-machine path planning and task allocation.

1.2. Key technologies of the unmanned farm

1.2.1. The state of research on path planning

Path planning technology is one of the key technologies for CO-

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MAMSY. An efficient, reasonable, and reliable working path can avoid path repetition, shorten the moving distance, and improve work efficiency. Path planning includes point-to-point path planning (P-PPP) and complete coverage path planning (CCPP). Research into P-PPP methods is relatively mature. Related research, such as path planning methods based on environmental maps, hybrid path planning methods, dynamic path planning methods, and optimization of path planning methods, continues to emerge and accelerate the development of related fields (He et al., 2023; Khajepour et al., 2020). In the field of agricultural machinery, these methods are mainly used for planning transfer paths between fields and local obstacle avoidance paths (Wang et al., 2022a, Wang et al., 2022b; Zhao et al., 2023).

(1) Point-to-point path planning

Habib et al. (2016) proposed a method that combines Voronoi Diagram (VD) and the Modified Ant Colony Optimization (M-ACO) algorithm for path planning of mobile robots. The method generated edges and vertices in the obstacle-filled space using VD, selected the nodes (the vertices generated by VD) using M-ACO, and then obtained the shortest path using P-PPP. Das et al. (2016) proposed a multi-robot optimal path planning method that combined improved particle swarm optimization (IPSO) with an improved gravitational search algorithm (IGSA). The proposed approach embedded the social essence of IPSO with the motion mechanism of IGSA and optimized the number of turns and arrival time. Liang et al. (2021) proposed a multi-machine dynamic path planning method for wheeled autonomous tractors. The method used an Improved Iterative Greedy (IIG) algorithm with total completion time as the comprehensive optimization objective. The optimization algorithm presented a remarkable performance in uncertain scenarios. Li et al. (2022) proposed a Dynamic Chaotic Ant Colony Optimization (DCACO) algorithm to address the problems of traditional Ant Colony Optimization (ACO) algorithms in P-PPP, such as long time consumption, slow convergence speed, and easily falling into local optimums. The cosine annealing strategy was used to improve the expectation heuristic factor and balance the global search ability of the algorithm to achieve better results.

(2) Complete coverage path planning

CCPP is mainly used for path planning in harvesting operations by harvesters, spraying operations by drones, and tillage and fertilizing operations by agricultural machinery within a field. There are various types of CCPP methods, including simple path planning methods, optimized path planning methods, and hybrid path planning methods (Wang et al., 2022a, Wang et al., 2022b). The most commonly used methods for simple path planning are nested and reciprocating. Optimized path planning methods have been proposed due to the high repetition rate and omission rate of simple path planning (Jeon et al., 2021; Wang et al., 2023). Subsequently, researchers have combined multiple path-planning algorithms to propose hybrid path-planning methods (Zhu et al., 2019).

Lei et al. (2022) proposed a new type of deep learning-based CCPP method. A feature-learning-enabled fully convolutional deep neural network (FCNN) model supporting feature learning was developed to identify edges in the exploration workspace. In path planning using deep learning, the deep learning model was combined with the start and end positions of the vehicle to generate CCPP paths. Chen et al. (2022) proposed a mixed-rules path planning algorithm based on the simulated annealing algorithm. An improved Douglas-Peucker fitting algorithm was used to handle the block boundary and obstacle boundary. In the traversal order of agricultural machinery, the crop rows were decomposed into units, and the optimal units were solved by the simulated annealing algorithm to form a complete coverage path. Edwards et al. (2017) used the optimized infield path planner developed at Aarhus University in Denmark for route planning, which shortened the path length and reduced the traffic intensity. Santoro et al. (2017) proposed a mathematical model to solve the mechanized harvesting route planning problem. This mathematical model minimized the time spent maneuvering the harvesting machine to cover all sugarcane rows, reducing fuel and labor costs.

1.2.2. The state of research on task allocation

Multi-machine task allocation technology is one of the important technologies for CO-MAMSY. It is also the core technology for improving the efficiency of CO-MAMSY. Most of the research on multi-machine task allocation has been at the theoretical research stage, with less application to actual production (Cao et al., 2021; Li et al., 2023). Currently, research in this area concentrates on UAVs, robots, and other areas, with less emphasis on agricultural machinery (Josh et al., 2020; Rahim et al., 2023). Recent research has focused on task allocation in a dynamic environment and multi-objective task optimization.

Liu et al. (2021) constructed an optimization model to maximize the net income of agricultural machinery households and used the particle swarm algorithm to solve it. The path allocation of agricultural machinery and cross-regional operations were achieved. D'Urso et al. (2018) proposed an algorithm based on dynamic programming to solve small-scale agricultural instance problems, and an approximate anytime algorithm based on the branch and bound approach to solve large-scale problem instance problems with numerous machines (robots). The simulation results of the proposed algorithm were applied to agricultural operations such as paint spraying, broadcast spraying, and manure application. Conesa-Munoz et al. (2016) studied path planning for herbicide spraying at specific locations and proposed an agricultural machinery task allocation algorithm based on a simulated annealing algorithm. Taking into account different criteria, different vehicle characteristics, field variability, tank refilling possibility, and other conditions, the model solved the agricultural machinery task allocation problem, and determined the optimal coverage sequence of farmland, the trajectory of agricultural machinery, and the optimal transition time for the filling of agricultural products. Seyyedhasani and Dvorak (2018) studied the problem of dynamic task allocation, which may require the redistribution and reordering of paths in agricultural field operations. The improved Clarke-Wright saving algorithm was used as the heuristic process to generate the initial solution. The Tabu search algorithm was used as the *meta*-heuristic process to obtain the optimal solution. Although this method can dynamically update the working path of the machines, if the dynamic deviation has not been significant, it may provide no improvement or even worse solutions. Bochtis and Sorensen (2009), Bochtis and Sorensen (2010) proposed the concepts of service units (SU) and primary units (PU) for the vehicle routing problem with time windows in cooperating heterogeneous agricultural machines. In the case of field operations involving cooperating machines, an SU or a team of SUs was required to fulfill a request for on-site services from PUs. A theoretical method was used to solve the planning and scheduling tasks of the SUs and to obtain paths for the SUs.

1.2.3. Summary of key technologies

In summary, the existing path planning and task allocation methods have some drawbacks, mainly including: (1) Lack of a reasonable and effective scheduling scheme when there is a shortage of agricultural machinery. (2) Lack of a fast inter-field path planning method and a low-cost intra-field coverage path planning method. (3) Lack of an effective whole-process path planning scheme for unmanned farms. In general, complete coverage path planning is the focus of research in a single field. Global path planning and task allocation for multi-agricultural machines with multiple fields are rarely considered. Unfortunately, the research focus of multi-machine collaborative operations is on task allocation and whole-process path planning. The problem of task allocation and path planning for multi-field applications is not considered in the actual situation of large farms, which makes it difficult to meet the production needs of unmanned intelligent farms.

1.3. Research objectives

To address the above issues, this paper conducts research on path planning and task allocation in the CO-MAMSY. Multi-machine collaborative path planning and task allocation are key issues in CO-MAMSY. It is also a complex integrated scheduling problem involving path planning, task allocation, and environment map building. It must ensure the rationalization of multi-machine task allocation, the minimization of path costs, and the maximization of farm profit. In this study, a collaborative navigation and scheduling method for multiple agricultural machines was proposed to provide map services and algorithmic decision support for multi-machine application scenarios and to solve the problem of path planning and task allocation for the CO-MAMSY.

The multiple machines collaborative operation scheduling module was developed. The simple path planning methods were utilized to plan the intra-field complete coverage path. Additionally, an improved Dijkstra algorithm was proposed based on the priority queue to plan the inter-field transfer path. Considering the field coverage path and the transfer path, the optimized ant colony algorithm was used to assign the tasks of the agricultural machines and finally to find the whole-process working path from the departure of the hangar to the return of the hangar. The research provides a theoretical basis for solving the scheduling management of group coordination operations in farmland environments and offers a technical scheme for promoting the construction of unmanned farms.

2. Problem description

To effectively plan paths and allocate tasks for CO-MAMSY, it is essential to establish a clear mapping relationship between multiple agricultural machines and multiple farmland plots (Cao et al., 2023). Due to the small-scale and scattered nature of most farmlands in China, it is common to find multiple plots on a single farm. As a result, there is a large amount of farmland that needs to be serviced by multiple machines. The farm adopts standardized agricultural machinery management, which includes unified parking and storage, dispatch and management, maintenance, and repair. The implementation of unified agricultural machinery allocation can increase the work area, alleviate difficulties such as saturation of agricultural machinery quantity and insufficient workload, and maximize farm profitability.

Based on information such as farm roads, task unit (plot) locations, and agricultural machinery, an appropriate task allocation strategy can be developed to minimize path costs and plan convenient working paths. Each task unit can be considered as a service point. This model takes into account both the transfer path between the hangar and the service location and the coverage path within the service point. All agricultural machines leave the farm's hangar, carry out agricultural activities according to the scheduled working path and assigned task units, and then return to the hangar. Finally, scheduling and management of multiple agricultural machinery collaborative operations within the farm will be achieved.

To simplify the model calculation for the complex and changeable working environment of the farm field, the basic assumptions for the path planning and task allocation problem for multi-machine collaboration were defined as follows:

(1) In task allocation, one task unit is only operated by one agricultural machine.
(2) All agricultural machines start from the hangar and return to the hangar after the operation is completed. Only one hangar can be built on the same farm.
(3) All work units have no obstacles, the equipment operation parameters are the same, and there are no faults in the operation process.
(4) For safety reasons, all roads can be used by two agricultural machines at the same time in order to avoid collisions during the transport of agricultural machines.
(5) Every agricultural machine has enough fuel to accomplish the task.
(6) When agricultural machinery is completely covered in the field, it is not affected by external elements during the driving process and runs at a constant speed.
(7) Agricultural machinery can only enter and leave the farmland at the entry and exit points.

3. Path planning and task allocation for multi-machine collaboration

3.1. Path planning and task allocation framework

Path planning and task allocation in the same type of multi-agricultural machine collaboration consist of four parts: constructing an environmental map, transferring path planning between fields, completing coverage path planning inside the field, and allocating multi-machine tasks. Fig. 1 depicts the scheme map for multi-agricultural machine collaboration.

(1) Constructing an environmental map

Constructing an environmental map is an important part of path planning (Han et al., 2023). According to known environmental map information, a specific mathematical model is utilized to depict the working environment as well as the process of transforming the real environment into map feature information that agricultural machinery can recognize and easily store. The Zhuozhou Experimental Farm of China Agricultural University (Zhuozhou City, Hebei Province, China) served as the study's foundation, and the environmental map was generated using the topological map method.

(2) Transfer path planning between fields

Path planning in the traditional sense refers to P-PPP, which means that agricultural machinery plans the shortest path from the start point to the endpoint without collision (Zhang et al., 2022).

The field transfer path planning in this study mainly includes the path planning between the hangar location point and the boundary point of the work plot, the path planning between the boundary point of work plot A, and work plot B, which is the path planning between the two points. The path planning between the hangar location point and the work plot boundary point primarily includes the path of the agricultural machinery from the hangar to the work plot, where the start point is the hangar location point and the endpoint is the work plot's entry point. The path of agricultural machinery returning from the farm to the hangar, where the start point is the farm exit point and the endpoint is the hangar location point.

The path of the agricultural machinery from the present work plot to the next work plot is planned between the border points of work plot A and work plot B. The start point is the current work plot's exit point, and the endpoint is the following work plot's entry point. The improved Dijkstra algorithm based on priority queues was used to design the field transfer path and to determine the shortest path for each work based on the path cost minimization concept.

(3) Complete coverage path planning within the field

The path planning within the field in this study is CCPP, where the working path needs to scan and traverse the entire work area, eventually forming a continuous path that passes through all regions within the work area from the start position (Qin et al., 2023; Selek et al., 2022).

The start and end locations of the path are the plot's boundary

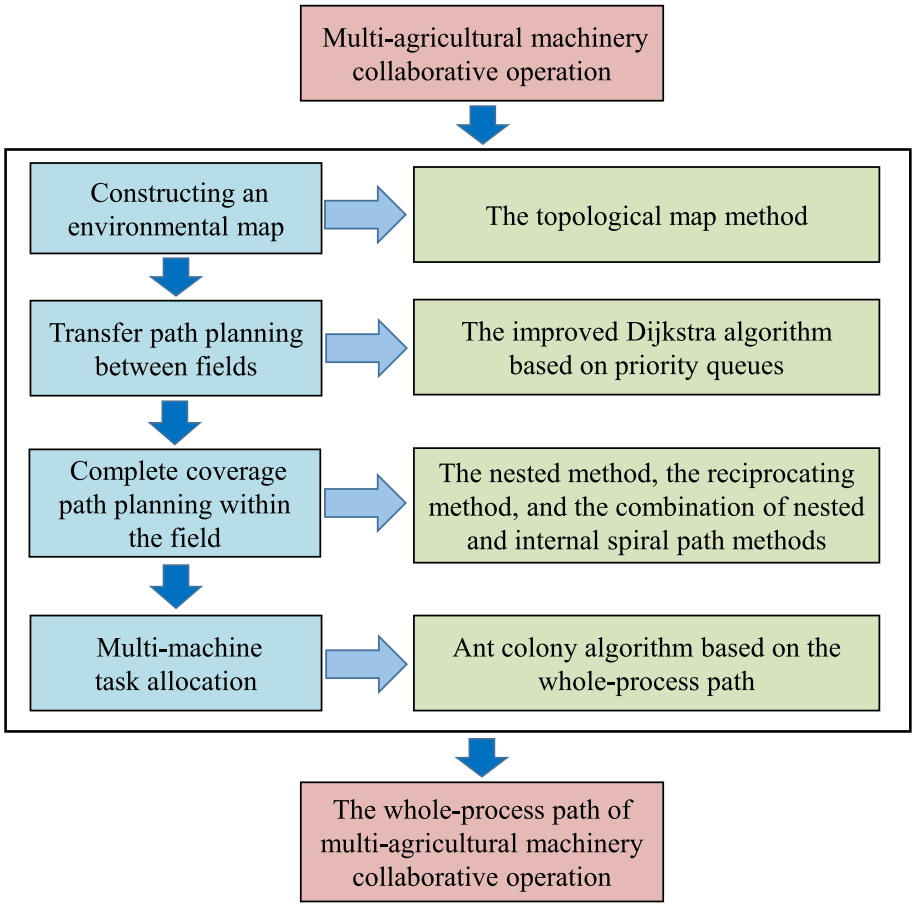


Fig. 1. Multi-agricultural machine path planning and task allocation scheme map.

points, and they will not be the same. The start point of the plot is determined by the preceding segment of the transfer path and is usually the closest boundary point from the path to the last work location. The endpoint of the plot is determined by the coverage path, which is usually the closest boundary point to the complete coverage path. The nested, reciprocating, and nested and internal spiral combined path planning methods were used to plan the field-covering path based on the shape of the farm plot.

(4) Multi-machine task allocation

Multi-machine task allocation in the unmanned farm refers to setting an evaluation index for multiple agricultural machines (multiple robots) and assigning a series of tasks to specific agricultural machines (robots) so that the agricultural machines (robots) have the highest returns while all tasks are completed (Korsah et al., 2013). To achieve task allocation for multi-agricultural machines, the ant colony algorithm based on the whole-process path was used, which combined the distance of the field transfer path and the distance of the complete coverage path with the goal of the lowest path cost.

3.2. Method for constructing an environment map

From October 20 to 22, 2022, environmental data was collected at Zhuozhou Experimental Farm for road information and in-field photos. Fig. 2 illustrates the use of RTK differential positioning technology to construct a ground base station and an electric vehicle carrying a GNSS mobile station to gather latitude and longitude position information of the farm road. Additionally, a small UAV was used to capture picture data in the field.

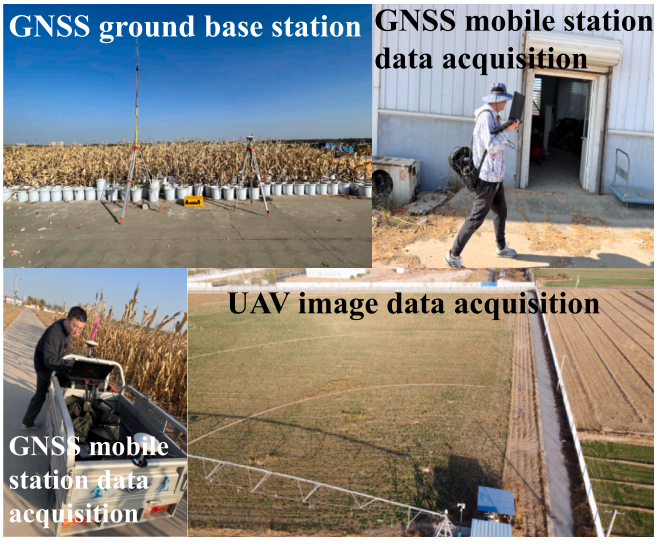


Fig. 2. Diagram of environmental data collecting.

This study generated an electronic map of the agricultural environment's topography based on information on road distribution and field crops at Zhuozhou Experimental Farm to reflect the farm's overall layout. The farm had twenty experimental plots, including eleven farmland plots, two greenhouse plots, one paddy field plot, one orchard plot, one cotton plot, one vegetable garden plot, one garden plot, one hangar plot, and one wasteland plot. The boundary points' latitude and longitude of each plot were acquired and digitally marked. A straight

line was utilized to link the boundary points and establish the borders of each plot. The topological map technique was used to generate an undirected connection map. Fig. 3 depicts the completed electronic map.

3.3. Transfer paths planning based on improved Dijkstra algorithm

The Dijkstra algorithm is based on employing the breadth-first search strategy to identify all the nearby vertices of the current vertex, extending layer by layer, until the shortest path to the endpoint is found (Zhou & Huang, 2022). In general, an S set (the point that has discovered the shortest path) and a U set (the point that has not yet discovered the shortest way) are required. All the points in U are explored, the shortest path to a point is found each time, and a point is selected from U to join S until U is an empty set. The algorithm's temporal complexity is O (n^2).

The traditional Dijkstra algorithm spends more time finding the edge of the adjacency matrix and searching for the current shortest path. To improve the response speed, the Dijkstra algorithm based on priority queues was introduced. The priority queue is ordered by the shortest distance value. The lower the value, the higher the priority. The shortest distance value in the queue can be accessed quickly by locating the node. The improved Dijkstra algorithm, which is based on the priority queue and minimizes computer complexity while enhancing algorithm performance, is seen schematically in Fig. 4.

Fig. 5 depicts the flow of improved Dijkstra's algorithm based on the priority queue, and the main notion of the method is as follows:

(1) Initializing the node, giving each node a path weight, and setting the path value of the start point (source point) to 0.
(2) Establishing a priority queue. The queue element is used to record the number of nodes and the shortest path value of nodes and then push the start point into the queue.
(3) If the queue is not empty, carry out the following actions:
    (a) x equals the top node, and w equals the top node's shortest path value, v is the current node.
    (b) After traversing all of the x's nearby points, if the shortest path value of node is smaller than the current value of node v, we update v and add v to the queue.
(4) Traversal is completed.

Using a priority queue has a temporal complexity of O (nlogn).

3.4. Complete coverage path planning method within the field

Path planning methods typically utilized in the field include the external spiral path, the internal spiral path, the reciprocating path, the nested path, and so on. In this work, three methods were used to achieve complete coverage path planning in the field: nested, reciprocating, and a combination of nested and internal spiral. The agricultural plots of Zhuozhou Experimental Farm have a somewhat regular shape, and the boundary points are joined to form a closed polygon based on the coordinates of the plots' border points. Based on the head-turn pattern and operational width, a CCPP approach is employed to construct a complete operational path inside the field.

3.4.1. The nested path method

The nested path method is based on the inward and outward methods, and the two adjacent paths are separated by multiple working widths so that the machine moves back and forth until the entire area has been traversed. This method avoids turning narrow bends and is simple to use. However, there are many empty rows, a lot of back and forth, and it is easy to miss tillage.

The flowchart of the algorithm for the nested path method is shown in Fig. 6(a). The following are the specific implementation steps:

(1) Generating the plot. Using the coordinates of the plot's border points, generate plot boundary polygons to produce plot regions.
(2) Dividing the turnaround area. Two turning regions and the agricultural machinery work area are split based on the agricultural machinery's minimum turning radius and operating breadth.

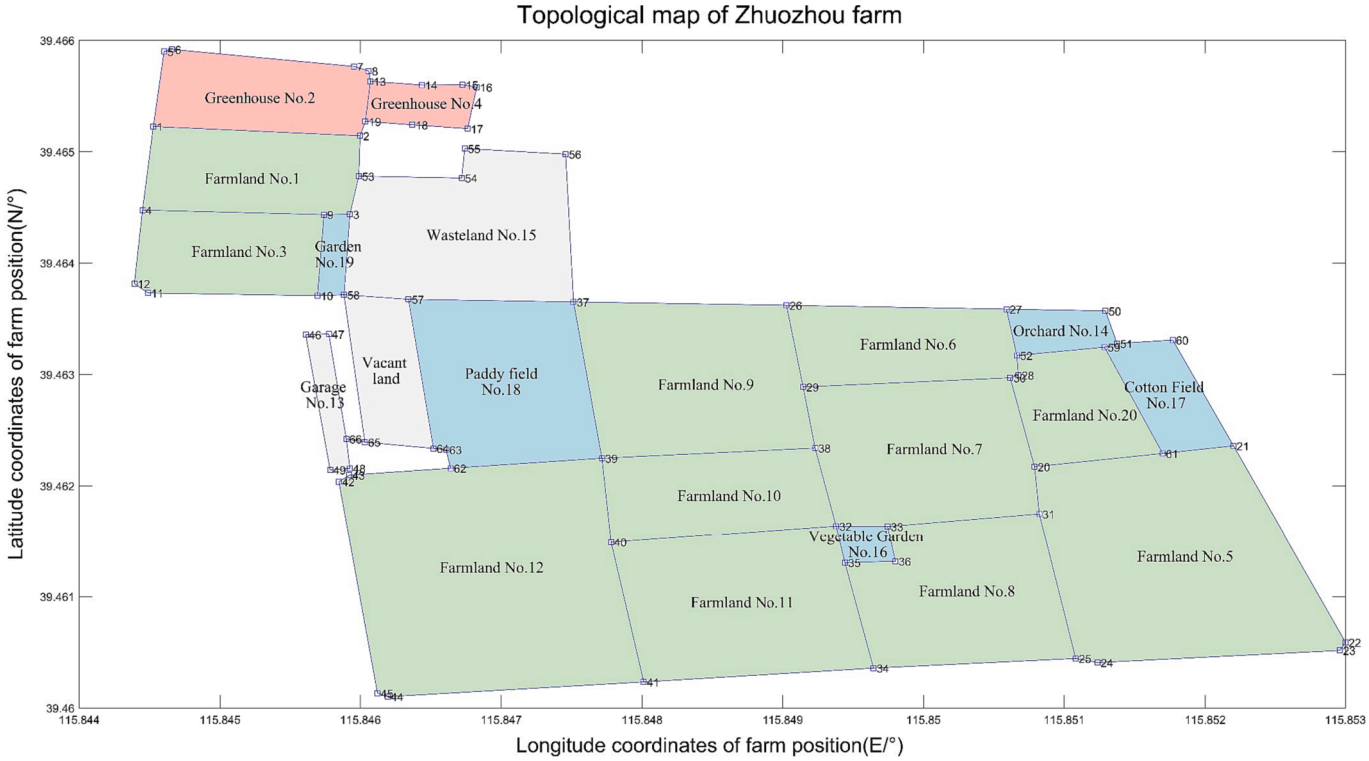


Fig. 3. Electronic map of the Zhuozhou Experimental Farm.

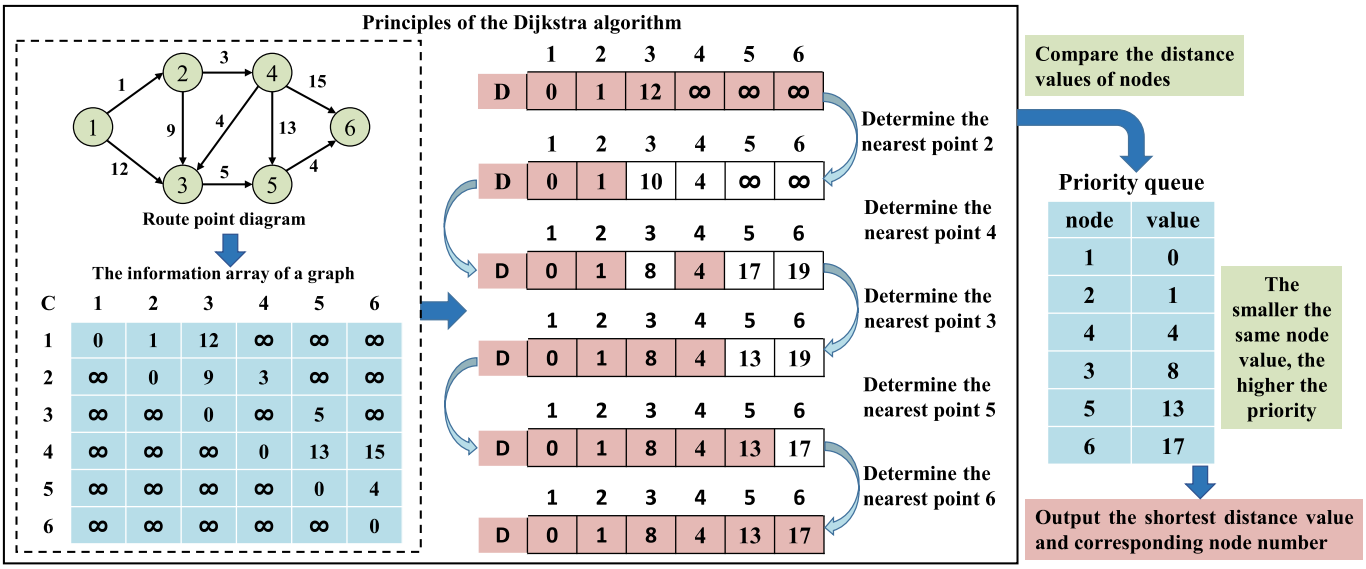


Fig. 4. Diagram of improved Dijkstra algorithm based on priority queue.

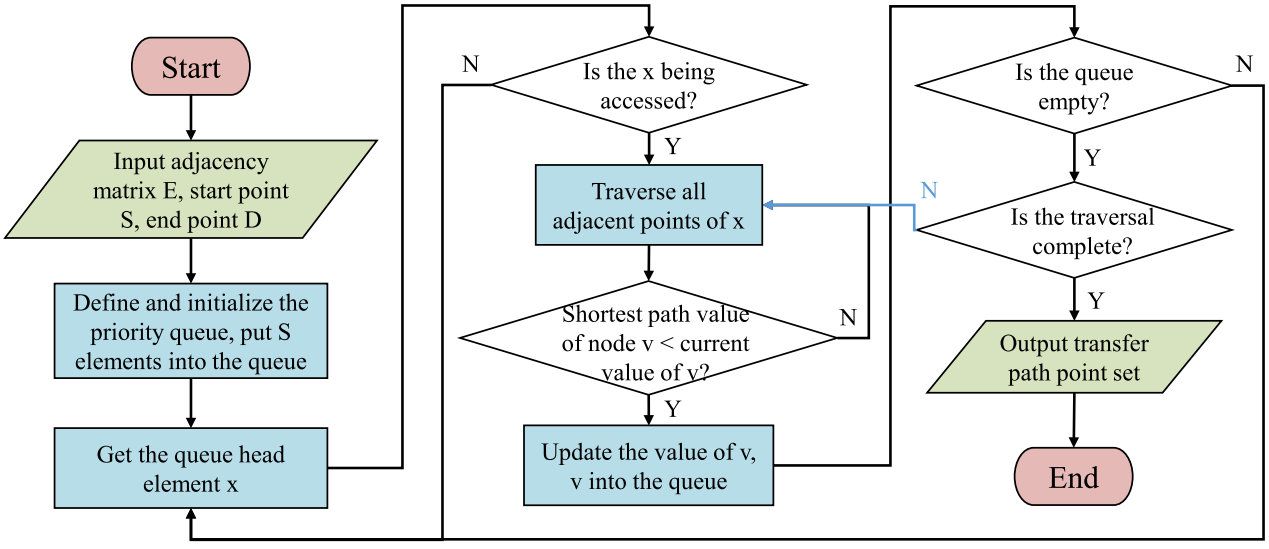


Fig. 5. Flowchart of improved Dijkstra algorithm based on priority queue.

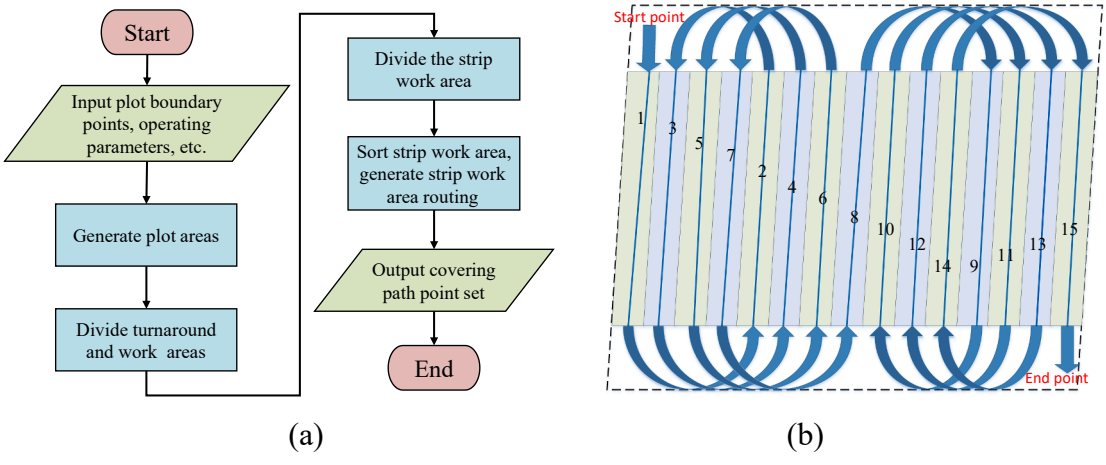


Fig. 6. Flowchart and schematic diagram of the nested path method.

(3) Dividing the strip work area. The overall width of the strip is determined by the distance between the two border points on the non-parallel side, and the width of each strip is determined by the width of the agricultural machinery operation, which yields the number of strips. The coordinates of each strip's four boundary points may be retrieved, and the border of each strip can be drawn using the coordinates of the boundary points, the width, and the number of strips.
(4) Sorting the strip work area. All strips are distributed evenly into N cells, with the number of strips per cell n and the number of cells N computed using formulas (1) and (2):

$$n = \left\lceil 2 \times \frac{r}{w} \right\rceil \times 2 + 1 \tag{1}$$

$$\mathrm{N} = \left\lfloor \frac{num}{n} \right\rfloor - 1 \tag{2}$$

where r represents the minimum turning radius of agricultural machinery, w represents the width of agricultural machinery operation, and num represents the number of strip work areas. The symbol ⌈ ⌉ represents the smallest integer greater than or equal to the given expression, while ⌊ ⌋ represents the largest integer less than or equal to the given expression. Formula (3) is used to compute the operation strip number $q[i]$:

$$q[i] = \begin{cases} \left\lfloor \frac{i-1}{n} \right\rfloor * n + mod[mod(i,n)+1,2] * \left\lceil \frac{n}{2} \right\rceil + \left\lceil \frac{mod(i,n)}{2} \right\rceil, i \le n*N \\ n*N + mod[(i-n*N),2] * \left\lceil \frac{num-n*N}{2} \right\rceil + \left\lfloor \frac{i-n*N}{2} \right\rfloor, i > n*N \end{cases} \tag{3}$$

In Formula (3), the mod is the modular operator, and i is the number of the strip work area's start section, sorted from left to right.

(5) Generating the turning path. At the moment, there are three types of ground turning methods: U type, T type, and Ω type. T-shaped and Ω-shaped turning is difficult and takes a long time, so U-shaped turning is selected.
(6) Outputting the set of working path points.

The schematic diagram of the nested path method is shown in Fig. 6 (b).

3.4.2. The reciprocating path method

The reciprocating method involves the machine entering the plot from one side and turning to the adjacent second row, so that the machine loops back and forth until the entire area has been traversed. The advantages of this method are fewer empty rows and high time utilization. However, it is not easy to maneuver the large machine around a narrow bend.

The flowchart of the algorithm for the reciprocating path method is shown in Fig. 7(a). The following are the specific implementation steps:

(1) Generating the work plot. A polygon is constructed to establish the operating region based on the coordinates of the plot's border points.
(2) Generating working path points. Rotate the work area, search for the best coverage angle, and generate coverage path points based on the working width.
(3) Planning the path. Reciprocal traversal of all path points to generate complete coverage paths based on the location of the entry and exit points.

The schematic diagram of the reciprocating path method is shown in Fig. 7(b).

3.4.3. Combination of nested and internal spiral path methods

The path of the internal spiral method gradually moves from the periphery to the center and finally covers the entire area. The advantage of this method is that the operation is simple with a small turning radius, but the number of turns is greater and the practical application is less. This research proposes a combination of nested and internal spiral path methods. The internal spiral method is used to cover the peripheral path of the plot, and then the nested method is used to cover the remainder.

The flowchart of the algorithm for the combination of nested and internal spiral path methods is shown in Fig. 8(a). The following are the specific implementation steps:

(1) Generating the work area. A polygon is formed based on the block's border point coordinates, and the polygon work area is built in the vertical direction.
(2) The peripheral path planning. The edge of the biggest polygon is the outermost path. The paths are progressively planned from the outside in, with each pair of four edges forming a group, the distance between the groups varying with the working width, and

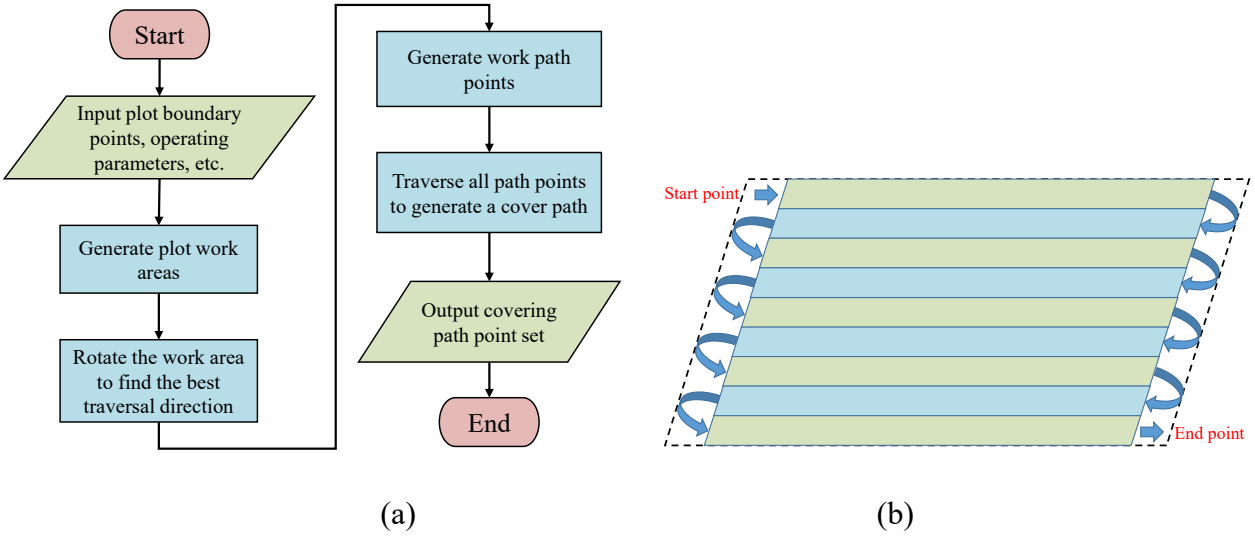


Fig. 7. Flowchart and schematic diagram of the reciprocating path method.

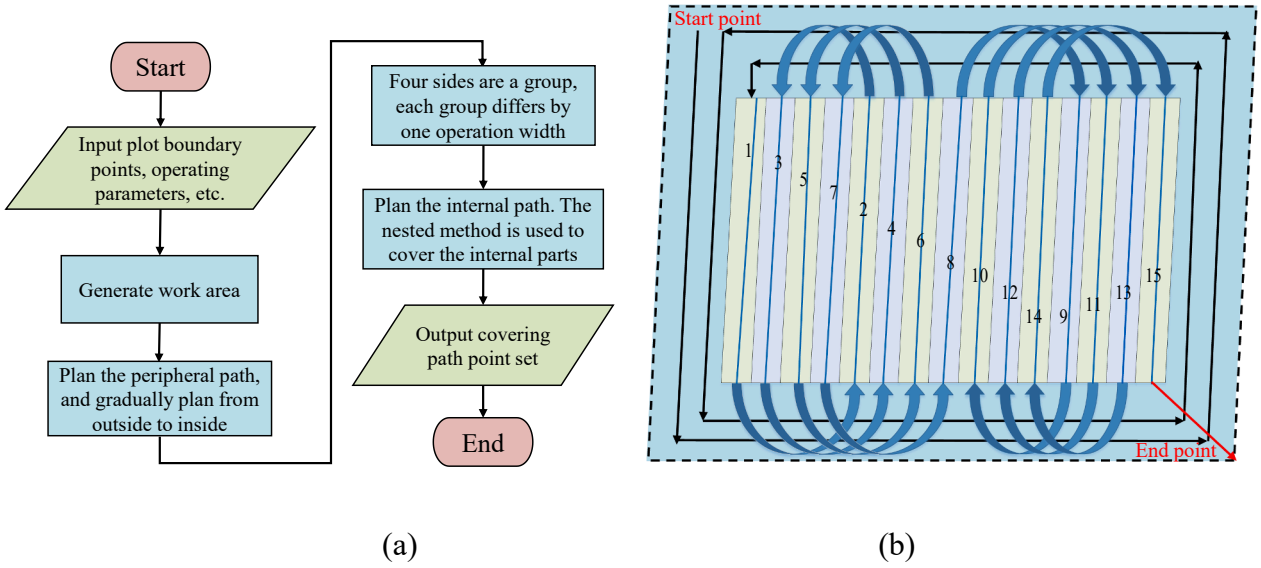


Fig. 8. Flowchart and schematic diagram of the combination of nested and internal spiral path methods.

each group differing by one working width until it reaches the turning radius of agricultural machinery.

(3) The internal path planning. The nested method is used to cover the internal parts.

The schematic diagram of the combination of nested and internal spiral path methods is shown in Fig. 8(b).

3.5. The task allocation method based on whole-process path

The Ant Colony Algorithm is a *meta*-heuristic optimization algorithm based on the simulation of the behavior of ants foraging for food. It has the qualities of a quick solution and can identify the global optimal solution while keeping the system's cost to a minimum (Li et al., 2017).

This paper suggested the ant colony method based on the whole-process path to achieve task allocation for agricultural machinery to reduce path costs. The distance of the field transfer path and the distance of the complete coverage path were examined to solve the problem of minimizing the distance when distributing agricultural machinery tasks. The concentration of pheromones emitted by ants as they move is utilized to determine the ideal task sequence (the best combination of paths). The number of tasks is inversely linked to pheromone concentration. The better the task sequence, the higher the pheromone concentration. The ants will choose a path with a higher pheromone concentration, leading to an increase in pheromone concentration on the best task sequence and a gradual decrease in pheromone concentration on other task sequences, resulting in a positive feedback system. Eventually, the ant colony works together to choose the ideal task sequence.

The flowchart of the ant colony algorithm based on the whole-process path is shown in Fig. 9. Each iteration of the ant colony algorithm produces the optimal allocation strategy under the current iteration, the current local optimal solution (Yun et al., 2022). Increasing the number of iterations will bring the local optimal solution closer to the global optimal solution, but will increase the algorithm's computing and time cost. As a result, the appropriate number of iterations must be determined to keep the algorithm's time cost to be tolerable. In this

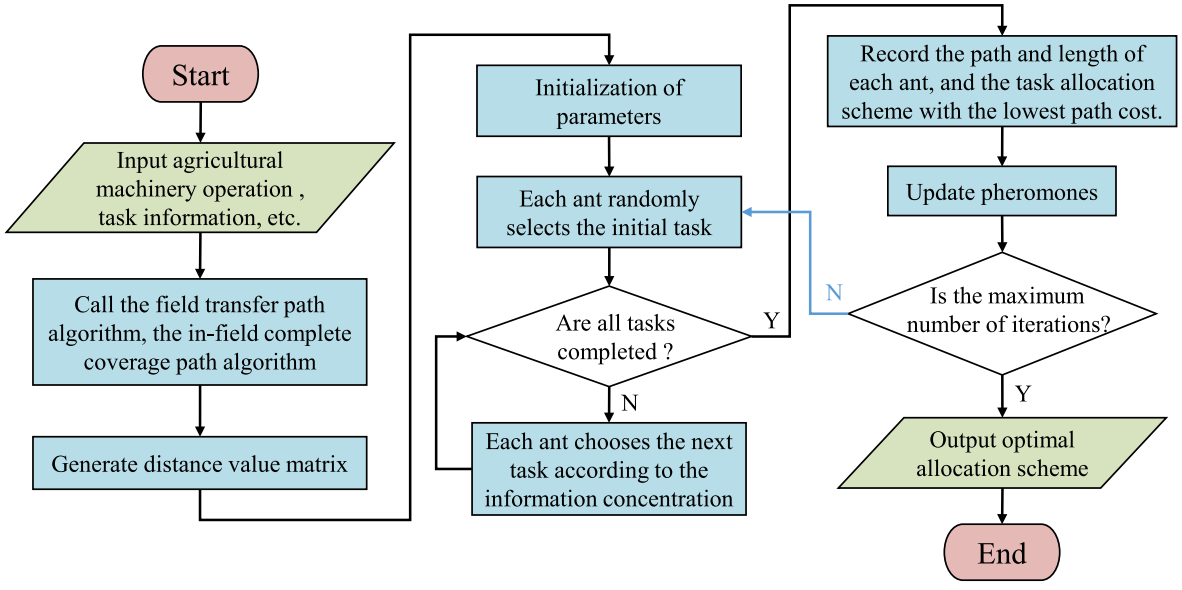


Fig. 9. Flowchart of the whole-process path based on ant colony algorithm.

research, the number of iterations is set to 100, and the number of ant colonies is set to 100.

The specific implementation steps of the ant colony algorithm are as follows:

(1) After creating the distance matrix, the transfer path planning algorithm is used to estimate the transfer distance, and the complete coverage path planning algorithm is used to determine the coverage distance.
(2) Setting and initializing the ant colony algorithm parameters and pheromone matrix, including the pheromone concentration, the path of the ants, the total length of the tasks, and the total time of the tasks.
(3) If the number of iterations is less than the set 100 and all tasks are not finished, perform the following operations:
   (a) The ant's allocation path is constructed using the task allocation function, and the ant's path length is determined using the distance matrix and the ant's path.
   (b) Using the number of ants (+1) and the number of iterations (+1) to recalculate the pheromone.
(4) Determining the ideal path length by comparing all ants' paths, and then outputting the results of the optimal task allocation sequence.

4. Algorithm simulation experiments and data analysis

4.1. Simulation conditions and parameter settings

Simulation tests were conducted to verify the feasibility and effectiveness of the algorithm under study for path planning and task allocation in multi-agricultural machine collaboration. The tests used real farmland data and field operation parameters based on the basic assumptions in Section 2. The simulation experiment was conducted on a computer with the Windows 10 operating system, an Intel(R) Core(TM) i7-8750H at 2.20 GHz, an NVIDIA GeForce GTX 1060 GPU, and 16 GB of RAM. MATLAB 2020 (MathWorks, USA) was used to conduct the simulation algorithm.

The electronic topographic map uses the electronic map of Zhuozhou Experimental Farm established in this study, as shown in Fig. 3. The rectangular squares indicate road junctions and map border locations. The solid lines linking the nodes depict the roadways on which agricultural machinery may travel as well as the plot boundary lines.

There are 66 nodes on the map, each marked with a number, and the weight of each edge is calculated based on the latitude and longitude coordinates of the node. The number of task plots for the simulation trial was 11, based on the number of arable fields, namely farm plots 1, 3, 5–12, and 20. The latitude and longitude coordinates of some nodes on the topographic map are shown in Table 1.

4.2. Verification test of the in-field path planning algorithm

The farmlands No. 1, No. 6, and No. 10 of the experimental farm were randomly selected. Each farmland's coverage path was designed using three methods: nested, reciprocating, and a combination of nested and internal spiral. The working width was 2 m, and the working area ratio was determined. The results of the associated parameters are shown in Table 2.

Table 2 shows that the reciprocating method's coverage area is greater than that of the nested method and the nested and internal spiral combination method, and its coverage index is higher than that of the other two ways. The nested method has an average coverage rate of 82.34 %, the nested and internal spiral combination method has an average coverage rate of 93.39 %, and the reciprocating method has an average coverage rate of 94.73 %. The coverage of the reciprocating method and the nested and internal spiral combination methods is greater than 92 %, but the average coverage of the reciprocating method is slightly higher, and it is used to plan the complete coverage path within the fields. The nested method has limited coverage because the polygon work area is built vertically and certain work areas are deleted, resulting in low coverage.

4.3. Verification test of the transfer path planning algorithm

To analyze the impact of different path lengths on the algorithm, different transfer distances were used to compare the running times of

Table 2
Results of in-field cover path experiments.

Farmland number	Cover method	Plot area (m^2)	Work area (m^2)	Work area ratio (%)
Farmland No.1	Nested path method	10334.6446	8389.3895	81.18
	Nested and internal spiral combination method		9715.2891	94.01
	Reciprocating path method		9869.0584	95.49
Farmland No.6	Nested path method	9951.5312	8519.4611	85.61
	Nested and internal spiral combination method		9951.5312	92.93
	Reciprocating path method		9281.6057	93.27
Farmland No.10	Nested path method	10972.0586	8966.6213	81.72
	Nested and internal spiral combination method		10229.4023	93.23
	Reciprocating path method		10471.6297	95.44

Table 1
Longitude and latitude coordinates of the nodes on the Zhuozhou Experimental Farm's topographic map.

Node number	Node latitude and longitude coordinate (X-axis/long.-E, Y-axis/lat.-N)	Node number	Node latitude and longitude coordinate (X-axis/long.-E, Y-axis/lat.-N)
1	(115.8445267,39.46522548)	16	(115.8468258,39.46557860)
2	(115.8459977,39.46514174)	17	(115.8467609,39.46520688)
3	(115.8459254,39.46443751)	18	(115.8463686,39.46524115)
4	(115.8444500,39.46447524)	19	(115.8460360,39.46527175)
5	(115.8446027,39.46589873)	20	(115.8507891,39.46217130)
6	(115.8446583,39.46591724)	21	(115.8522009,39.46235871)
7	(115.8459548,39.46576441)	22	(115.8529995,39.46058963)
8	(115.8460589,39.4657241)	23	(115.8529607,39.46051657)
9	(115.8457422,39.46443455)	24	(115.8512360,39.46040777)
10	(115.8456955,39.46370739)	25	(115.8510809,39.46044463)
11	(115.8444876,39.46373167)	26	(115.8490285,39.46362002)
12	(115.844392,39.46381526)	27	(115.8505897,39.46358523)
13	(115.8460711,39.46563119)	28	(115.8506741,39.46299345)
14	(115.8464358,39.46559784)	29	(115.8491467,39.46288730)
15	(115.8467253,39.46559931)	30	(115.8506163,39.46296956)

the two algorithms. The transfer paths from farmland No. 1 to farmland No. 9, farmland No. 6, farmland No. 7, and farmland No. 5 were selected. For different transfer distances, the performance of the Dijkstra algorithm before and after optimization is compared. Table 3 shows the comparative results.

The shortest transfer path length computed by the two methods is the same, but the improved Dijkstra algorithm based on the priority queue has a shorter execution time and better time performance than the classical Dijkstra algorithm, according to Table 3. It can be observed that as the length of the transmission path increases, the time advantage of the improved Dijkstra algorithm based on the priority queue becomes more apparent. The execution time is reduced by 2.24 %, 1.73 %, 3.15 %, and 4.59 %, respectively, compared to the classic Dijkstra algorithm. Due to the limited size of the farm, the maximum transfer distance is approximately 850 m, resulting in a maximum percentage reduction in execution time of 4.59 %. As the value of the transfer distance increases, so does the percentage of probable execution time.

4.4. Verification test of the multi-machine collaborative task allocation algorithm

Zhuozhou Experimental Farm has eleven plots of farmland and three agricultural machines. All three agricultural machines are situated in the garage at the latitude and longitude coordinates of (115.84603222330978 E, 39.46239269521264 N). To achieve multi-machine task allocation, the number of tasks should be greater than the number of machines by 3 and less than the total number of plots by 11, and set a task as a multiple of the number of agricultural machines. Therefore, the number of tasks is set to 4, 6, and 10, and the corresponding farm numbers for each task are shown in Table 4. The farm tasks are allocated according to the different numbers of tasks. The results are given in Table 5.

Table 5 shows that when the path length of the random sequence assigned by the task is compared to the optimal sequence, the optimal sequence's path cost is reduced. When compared to the random sequence, the path cost decreases by 14 %–33 %. It shows that the ant colony method based on the whole-process path, can successfully minimize the working path cost.

Table 3
Comparison of experimental results of transfer paths before and after optimization by Dijkstra algorithm.

Transfer path	Algorithm	Path length (m)	Average time of single execution (ms)	Percentage reduction in execution time (%)
From Farmland No.1 to Farmland No.9	Dijkstra algorithm	426.31	0.3386	2.24
	Dijkstra algorithm based on priority queue	426.31	0.3310	
	Transfer path points: 1,2,53,3,58,57,37			
From Farmland No.1 to Farmland No.6	Dijkstra algorithm	556.65	0.3423	1.37
	Dijkstra algorithm based on priority queue	556.65	0.3376	
	Transfer path points: 1,2,53,3,58,57,37,26			
From Farmland No.1 to Farmland No.7	Dijkstra algorithm	638.84	0.3494	3.15
	Dijkstra algorithm based on priority queue	638.84	0.3387	
	Transfer path points: 1,2,53,3,58,57,37,26,29			
From Farmland No.1 to Farmland No.5	Dijkstra algorithm	852.93	0.3680	4.59
	Dijkstra algorithm based on priority queue	852.93	0.3511	
	Transfer path points: 1,2,53,3,58,57,37,26,27,52,28,30,20			

Table 4
Farmland numbers corresponding to different numbers of tasks.

Task quantity	Farmland number
4	(1,6,10,11)
6	(1,3,5,6,10,11)
10	(1,3,5,6,7,8,9,10,11,12)

4.5. Simulation of the whole-process path planning

The above algorithms have been combined, and the integrated algorithms have been tested and confirmed to validate the applicability and practicality of the overall algorithm. The algorithm may plan the whole-process path of a single agricultural machine conducting multiple operation tasks or multiple agricultural machines of the same type conducting multiple operation tasks. By invoking the ant colony algorithm based on the whole-process operation and obtaining the respective task sequence, the algorithm can allocate the operational tasks. The improved Dijkstra algorithm based on the priority queue is used to design the field transfer path, and the shortest path trajectory point of the farm machinery transfer path from the start point to the endpoint is achieved. The reciprocating path algorithm is invoked, and the continuous path trajectory points in the present working area are traversed from the start point to accomplish complete coverage of the path planning within the field. Based on the work sequence assigned by the agricultural machine, the trajectory points of the transfer path and the trajectory points of the complete coverage path are merged into an array to achieve whole-process path planning.

A single agricultural machine conducting multiple operation tasks and multiple machines conducting multiple operation tasks were simulated and tested. Based on the electronic map of the Zhuozhou Experimental Farm shown in Fig. 3, whole-process path planning is carried out. The simulation test results for two cases are presented below.

Case1 Single agricultural machine conducting multiple operation tasks

Hangar No. 13 was selected for an agricultural machine, and the operation tasks were farmland No. 1, farmland No. 6, and farmland No. 10. In this method, the whole-process path was planned, and the agricultural machine returned to the hangar after completing all the operational tasks. Fig. 10 shows the expected whole-process path of the single-machine, multi-task operation.

The planned path shows that an agricultural machine starts from hangar No. 13, and the operation task order is farmland No. 10, farmland No. 6, and farmland No. 1, and finally returns to the hangar from farmland No. 1. Similarly, the agricultural machine is started from hangar No. 13, and the order of operation tasks is changed to farmland No. 1, farmland No. 6, farmland No. 10, and finally returned to the hangar by farmland No. 10. The only difference is whether the planned path is clockwise or anticlockwise. The path shown in Fig. 10 follows an anticlockwise direction and is divided into four primary segments. The first section, which began at the 66 points of hangar No. 13, continued through 65, 64, 63, and 62, and reaches entry point 39 of farmland No. 10. When the cover operation was finished, farmland No. 10′s exit point 32 departed. In the second section, the path began from farmland No. 10 point 32, traveled through 38 to farmland No. 6 entry point 29, executed a cover operation, and finally departed from farmland No. 6 exit point 26. In the third section, which began at farmland No. 6 point 26 and travels through 37, 57, 58, and 3 to farmland No. 1 entry point 2, the coverage operation was done, and farmland No. 1 exit point 4 was left. In the fourth section, the path started at point 4 of Farmland 1 and

Table 5
The results of task allocation with a variable number of tasks.

Quantity of tasks	Agricultural machinery number	Task number	Execution time (s)	Average cost of random sequence path (m)	Average cost of optimal sequence path (m)	Reduction rate of path cost (%)
4	1	(10,11)	0.2188	5980.38	3986.32	33.34
	2	(1)				
	3	(6)				
6	1	(5,6)	0.4022	12939.32	9896.11	23.52
	2	(10,11)				
	3	(1,3)				
10	1	(6,7,8,10,12)	0.9008	29614.90	25456.82	14.04
	2	(9,11)				
	3	(1,3,5)				

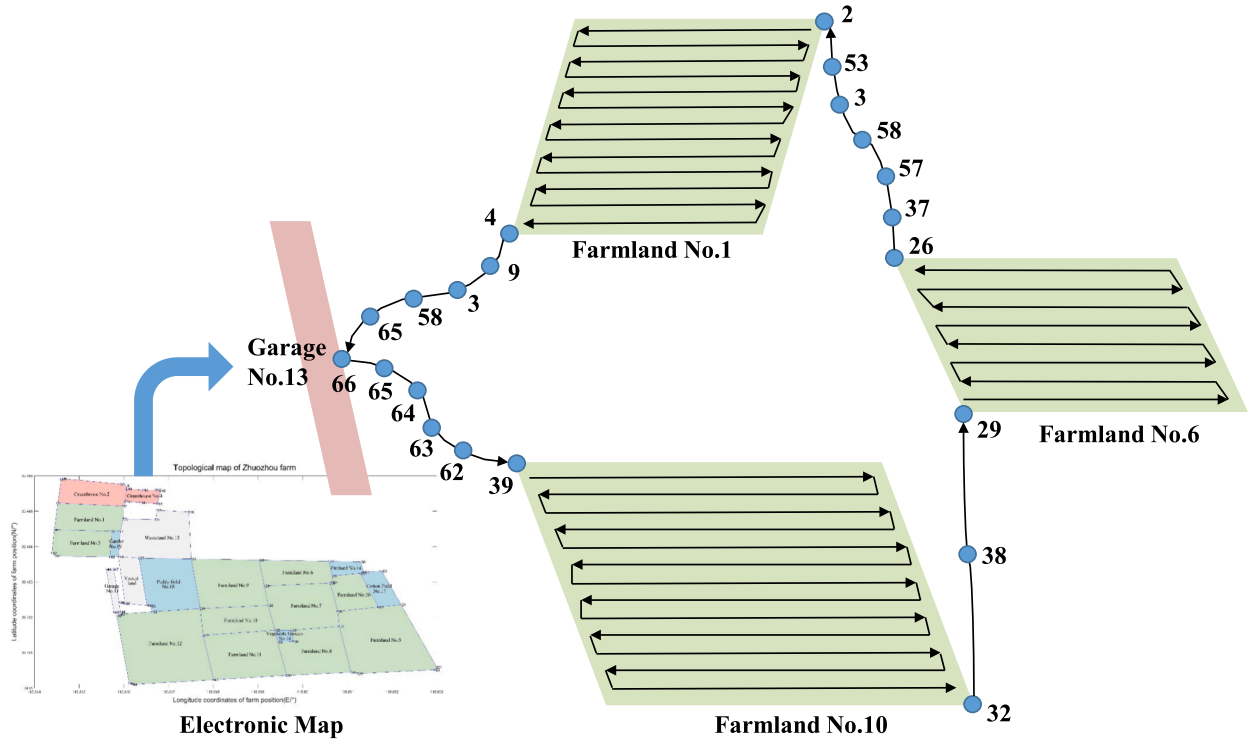


Fig. 10. The whole-process path diagram of a single machine multi-task operation.

returned to Hangar 13 via points 9, 3, 58, and 65 to reach point 66. The method fulfilled the path planning of a single agricultural machine, 'Hangar-Farmland-Hangar'.

Case2 Multiple agricultural machines conducting multiple operation tasks

The task assignments were farmland No. 1, farmland No. 3, farmland No. 6, farmland No. 7, farmland No. 10, and farmland No. 11, and three agricultural machines were selected from hangar No. 13. The agricultural machinery returned to the hangar after finishing all of the tasks. Fig. 11 depicts the whole-process path of the intended multi-machine, multi-task operation. The results of the whole-process path allocation for the multi-machine, multi-task operation are shown in Table 6.

Three agricultural machines were assigned to specific tasks, as shown in Fig. 11. The operation responsibilities of agricultural machine No. 1 included farmland No.1 and farmland No. 3, and the working paths were points 66, 65, 58, 10, 11, farmland No. 3, 9, 3, and farmland No.1, 2, 53, 3, 58, 65, 66. The operation task of the agricultural machine No.2 is farmland No. 6 and farmland No. 7, and the working path passes through points 66, 65, 64, 63, 62, 39, 38, 29, farmland No. 7, 31, 20, 30, 28, farmland No. 6, and 26, 29, 38, 39, 62, 63, 64, 65, 66. Farmland No.10 and farmland No.11 were used for the operational task of agricultural machine No.3, and the working path passes through points 66, 65, 64, 63, 62, 39, farmland No.10, 32, 40, farmland No.11, 34, 35, 32, 38, 39, 62, 63, 64, 65, 66. Table 6 shows the tasks assigned by the three agricultural machines, the order of the work plots, the working path points of the agricultural machines, and other information. The working path of agricultural machinery was composed of the transfer path and the coverage path, which realized the multi-task working path planning of multiple agricultural machines.

5. Conclusions

(1) A path planning and task allocation method for the collaboration of multiple agricultural machines on an unmanned farm was proposed. An electronic map of Zhuozhou Experimental Farm was constructed using the topological map method. To perform field transfer path planning, an improved Dijkstra algorithm based on a priority queue was presented. The basic CCPP approach was used in the field to design the complete coverage path. An ant colony method based on the whole working path was proposed to realize task allocation of multiple agricultural machines while aiming at the lowest path cost. By combining the path planning and task allocation methods, whole-process path

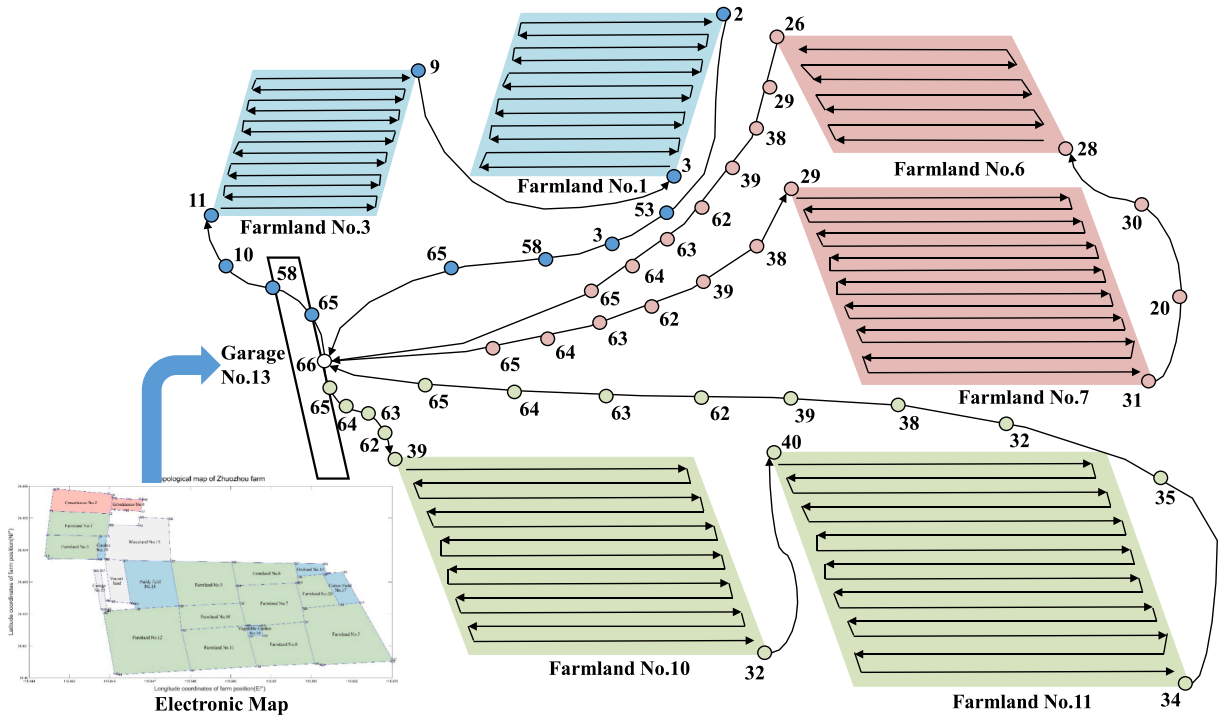


Fig. 11. The whole-process path diagram of the multi-machine multi-task operation.

Table 6
The results of the whole-process path allocation for the multi-machine, multi-task operation.

Task number	Agricultural machinery number	Description	Whole-process path				
			First part	Second part	Third part	Fourth part	Fifth part
(1,3)	1	Path points	66,65,58, 10,11	Farmland No. 3	9,3	Farmland No. 1	2,53,3,58,65,66
		Path types	transfer path	coverage path	transfer path	coverage path	transfer path
(6,7)	2	Path points	66,65,64, 63,62,39, 38,29	Farmland No. 7	31,20, 30,28	Farmland No. 6	26,29,38, 39,62,63, 64,65,66
		Path types	transfer path	coverage path	transfer path	coverage path	transfer path
(10,11)	3	Path points	66,65,64, 63,62,39	Farmland No. 10	32,40	Farmland No. 11	34,35,32,38,39,62,63,64,65,66
		Path types	transfer path	coverage path	transfer path	coverage path	transfer path

planning was achieved, and the working path of multi-agricultural machines of the same type was obtained.

(2) Based on the topological map of Zhuozhou Experimental Farm, the multi-agricultural collaborative path planning and task allocation algorithm was simulated using the MATLAB platform. The results show that the improved Dijkstra method based on priority queues can effectively reduce the running time. The reciprocating method had the highest coverage index, with an average coverage of 94.73 %. The proposed ant colony algorithm allocated tasks reasonably and effectively, reducing working path costs by 14 %–33 %. Combined with the above methods, the integrated algorithms realized the whole-process path planning of a single agricultural machine conducting multiple operation tasks or multiple agricultural machines of the same type conducting multiple operation tasks. It effectively solves the problem of path planning and task allocation for collaborative multi-agricultural machine operations in the complex working environment of regional farmland and provides a technical solution for promoting the construction of unmanned farms.

(3) In the next study, the obstacle problem within the farmland will be investigated in detail. The complete coverage path planning algorithm will be optimized. Currently, this research is still in the stage of algorithm simulation. The next step is to integrate the path planning and task allocation algorithms into the software management platform. After that, experiments will be conducted using unmanned farm application to evaluate and improve the algorithm's performance.

CRediT authorship contribution statement

Ning Wang: Conceptualization, Methodology, Writing – original draft. **Xi Yang:** Visualization, Data curation. **Tianhai Wang:** Formal analysis, Investigation. **Jianxing Xiao:** Investigation, Resources. **Man Zhang:** Writing – review & editing, Supervision. **Hao Wang:** Project administration. **Han Li:** Writing – review & editing, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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